Capstone: Draft the Problem Statement

Professional Certificate in Machine Learning and Artificial Intelligence

UC Berkeley

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An overview of the question to be solved in the capstone.

Problem Context:

- *My goal is to answer the core question: Given an audio voice sample, can I build a model to identify broad traits of the speaker such as gender and emotion similarly to human perception?*

Stakeholders & Value:

- Call centers and/or law enforcement call responders who struggle to efficiently triage incoming audio streams, often failing to detect high-intensity emotions like anger or distress in real time
- Hard-of-hearing neurodivergent individuals (such as those with autism spectrum conditions) who rely on written information to navigate conversations, media, or live environments

Decision Enabled:

- Operations managers or first responders could use automated predictions to dynamically prioritize distressed or escalated customer calls to the right department, especially when combined with Text-to-Speech (TTS)
- Assist in accessibility features by dynamically augmenting subtitles with rich descriptions of speaker traits and emotional tone

Identification of the type of data that will be used to solve the question.

I will be limiting my sources to North American English and plan to use industry standard structured audio-feature matrices extracted from audio datasets.

Possible Dataset Sources:

- The **RAVDESS** dataset (Ryerson Audio-Visual Database of Emotional Speech and Song) featuring professional North American actors.
- The **CREMA-D** dataset (Crowd-sourced Emotional Multimodal Actors Dataset), comprising clips from diverse actors speaking sentences across basic emotional states.

Data Characteristics:

- I intend for the raw data to be voice clips that are available freely. From these, I would have categorical vocal recordings mapped to distinct emotional states (happy, sad, angry, neutral) and actor demographic metadata. This is something I will have to build.

Feature Engineering Pipeline:

- Raw `.wav` files converted via `librosa` into fixed-length numerical feature arrays containing Mel-Frequency Cepstral Coefficients (MFCCs) and spectral centroids. These are the industry standard ways to turn voice clips into model-understandable data.

List of 1–3 techniques that could be used to answer the question identified.

Since there are multiple target variables here, I thought a progressive modeling approach starting with classical baselines and expanding to advanced classifiers would be helpful. I have read online that DNN is better for emotion recognition.

Baseline & Classical ML:

- Implementing K-Nearest Neighbors (KNN), Decision Trees, and Support Vector Machines (SVC) on scaled feature vectors. I plan to use this for basic gender identification

Advanced Architecture:

- Utilizing a Random Forest ensemble or a lightweight Neural Network as a stretch goal to capture complex non-linear acoustic relationships.

Rigorous Validation:

- Applying **Group** K-Fold Cross-Validation partitioned by Actor ID to prevent speaker bias and ensure generalization to unseen voices. In other words, I do not want the same speaker to appear multiple times in training and test data and have the model just memorize their voice rather than trends in the data.